REQUIREMENTS ENGINEERING DOCUMENT

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Table of Contents

1. Introduction
2. Stakeholder Identification and Analysis
3. Requirements Elicitation Plan
4. System Architecture (MVC Compliance)
5. Requirements Modelling and Analysis
6. Scenario and Use Case Development
7. Functional Requirements Specification
8. Non-Functional Requirements Specification
9. Technical Specifications
10. Requirements Engineering Roadmap Proposal
11. Presentation Readiness
12. **Conclusion**
13. **Appendices**

1. Introduction

1.1 Project Purpose

To design and implement a fully functional web-based system using the **Laravel framework** that digitalizes the recruitment process. The system enables employers to manage job listings (full CRUD) and allows job seekers to find and apply for positions, while following **MVC architecture** and Laravel best practices.

1.2 Problem Statement (As per exam requirement)

Manual or email-based job applications lack structure, tracking, and security. This system solves that by providing a centralized, role-based platform where all data operations (create, read, update, delete) are managed through secure authentication and validated forms.

1.3 Target Users (Five Stakeholders)

Stakeholder	Role in System
Job Seeker	Registers, searches jobs, applies, tracks status
Employer (Company/HR)	Posts jobs (CRUD), reviews applications, updates status
System Administrator	Manages users, monitors system, manages API keys
Government Labor Regulatory Body (e.g., MGLSD)	Receives anonymized reports, flags discriminatory jobs
Third-Party Service Provider (e.g., SMS, Email)	Provides notification delivery infrastructure

2. Stakeholder Identification and Analysis

The success of the Online Job Application System depends on understanding the diverse stakeholders who interact with, govern, and depend on the system. This section identifies and analyses the **five primary stakeholders**, their roles, needs, expectations, and constraints.

2.1 Stakeholder Register

S/N	Stakeholder	Category	Role in System
1	**Job Seeker**	End User	Registers, creates profile, searches jobs, submits applications, tracks application status, receives notifications.
2	**Employer (Company/HR)**	Business User	Registers company, posts job vacancies (full CRUD), reviews applications, shortlists candidates, updates application status, sends messages.
3	**System Administrator**	Technical Operator	Manages user accounts, monitors system performance and uptime, resolves disputes, maintains data integrity, manages API keys for third-party services.
4	**Government Labor Regulatory Body (e.g., MGLSD)**	Regulatory Authority	(Future/Compliance) Sets policies for fair employment, receives anonymized labor market data, audits job postings for discrimination, handles complaints.

5	**Third-Party Service Provider (e.g., SMS, Email)**	External Service	Provides notification delivery infrastructure (email, SMS) via API integration; required for system-to-user communication.

2.2 Stakeholder Needs and Expectations

2.2.1 Job Seeker

Job seekers require an intuitive, accessible, and fair platform to find and apply for jobs efficiently.

Their expectations include:

- Simple, quick registration and profile creation.
- Ability to upload CVs and cover letters in multiple formats (PDF, DOCX).
- Advanced job search with filters (location, salary range, job type, industry).
- Real-time application status tracking (pending, reviewed, shortlisted, rejected, hired).
- Receive instant email/SMS notifications for application updates.
- Secure storage of personal data and assurance of privacy.
- Ability to save jobs for later application.

2.2.2 Employer (Company/HR)

Employers need a reliable, efficient system to manage their entire recruitment workflow.

Their expectations include:

- Ability to post, edit, view, and delete job vacancies (full CRUD operations).
- A dashboard to view all applicants per job posting with sorting/filtering.
- Ability to update application status in bulk or individually.
- Direct messaging capability to contact shortlisted candidates.
- Analytics: number of views, applications per job, conversion rates.
- Export applications data (CSV/PDF) for offline record-keeping.
- Compliance with local labor laws through system prompts.

2.2.3 System Administrator

The administrator requires robust tools to keep the system running smoothly and securely.

Their expectations include:

- Complete user management (create, suspend, delete, role-change for seekers/employers).
- Content moderation (remove inappropriate job postings or fraudulent applications).

- System monitoring dashboard (uptime, response times, error logs, usage statistics).
- Ability to manage third-party service API keys (SMS, email).
- Backup and recovery mechanisms.
- Full audit trail of all critical actions (who did what and when).

2.2.4 Government Labor Regulatory Body (e.g., MGLSD)

The regulatory body needs the system to support fair labor practices and provide labor market insights.

Their expectations include:

- Access to anonymized, aggregated labor market data (e.g., hiring trends, salary ranges by sector).
- Ability to flag or request removal of discriminatory job postings.
- Receive reports on employer compliance with labor laws (e.g., minimum wage, contract types).
- A channel to receive and investigate complaints from job seekers.
- Assurance that the system operator (administrator) follows data protection laws.

2.2.5 Third-Party Service Provider (e.g., SMS, Email)

The service provider needs a reliable technical integration to deliver messages successfully.

Their expectations include:

- System must respect their API rate limits and terms of service.
- Proper formatting of requests (JSON/REST) with valid API keys.
- Handling of delivery receipts and webhook callbacks.
- Failure retry logic (e.g., retry failed email sends up to 3 times).
- Logging of all sent messages for billing and dispute resolution.

2.3 Stakeholder Power-Interest Matrix

Category	Stakeholders	Engagement Strategy
High Power, High Interest	Employers, Government Regulatory Body	**Manage closely:** Involve in compliance reviews, feature prioritization, and policy updates.
High Power, Low Interest	System Administrator (operational only)	**Keep satisfied:** Provide clear SLAs, monitoring tools, and escalation paths.
Low Power, High Interest	Job Seekers	**Keep informed:** Establish feedback channels (surveys, support email, FAQs).
Low Power, Low Interest	Third-Party Service Providers	**Monitor:** Ensure contractual SLAs are met; have backup providers.

2.4 Stakeholder Constraints (Uganda-Specific)

S/N	Stakeholder	Constraint	Uganda Specific Reality
1	Job Seeker	Limited internet access, low digital literacy, device cost	~47% rural population; mobile data costs remain high; many use feature phones only.
2	Employer (Company/HR)	Budget constraints for premium features, limited HR tech expertise	Many SMEs operate informally with no dedicated HR staff; expect free basic service.
3	System Administrator	Hosting infrastructure limitations, limited 24/7 staffing	Reliable cloud hosting (AWS/Azure) is costly; local hosting has power/internet risks.
4	Government Regulatory Body (MGLSD)	Limited enforcement capacity, slow policy updates	Labor inspectorate has few staff; online platform oversight is new and untested.
5	Third-Party Service Provider	Network reliability, cost per message	SMS delivery to rural networks may fail; email deliverability to free providers (Gmail/Yahoo) is unpredictable.

2.5 Assumptions and Limitations

2.5.1 Assumptions Made in this Analysis

- Job seekers and employers have access to basic smartphones or shared computers with intermittent internet.
- Employers will voluntarily comply with labor laws when using the system.
- The Government Regulatory Body will engage constructively if approached for data sharing.
- Third-party APIs (email/SMS) will remain available within agreed rate limits.

2.5.2 Limitations Acknowledged

- Direct interviews with government regulators were not conducted; analysis is based on published labor laws and policies.
- Power-interest classifications are estimates based on typical system roles.
- Advanced features (AI job matching, payment for job posts) are excluded due to scope and time.
- Refugee and stateless persons are not explicitly addressed in the current scope.

3. Requirements Elicitation Plan

This section outlines a structured, multi-technique approach for gathering requirements from all five identified stakeholders.

3.1 Elicitation Techniques

Technique	Target Stakeholders	Purpose	Timeline
Structured Interviews	Employers (HR managers), System Administrator candidates, Regulator representative	Capture strategic, operational, and compliance requirements.	Day 1–2
Focus Group Workshops	Job seekers (student/unemployed sample), Small business owners	Elicit usability expectations and pain points through group discussion.	Day 2–3
Surveys & Questionnaires	Job seekers (target 50+ responses)	Gather feature preferences and demographic data from a large user base.	Day 1–4
Document Analysis	Uganda Employment Act (2006), Data Protection Act (2019), existing job portal case studies	Extract regulatory and compliance requirements.	Day 1
Prototyping	Job seekers, Employers	Validate UI/UX through interactive Blade mockups before full development.	Day 5–7

3.2 Elicitation Schedule

Phase	Activities	Duration	Key Outputs
Phase 1: Preparation	Identify specific contacts, design survey instruments, collect legal documents	Day 1	Stakeholder contact list, interview guides, survey questionnaires
Phase 2: Primary Elicitation	Conduct interviews, distribute surveys, hold workshop	Day 2–3	Interview transcripts, workshop notes, raw survey data
Phase 3: Secondary Elicitation	Document analysis of laws and competitor portals	Day 2–3	Regulatory requirements matrix, best practices list
Phase 4: Validation	Prototyping sessions with sample users, review with regulator (if accessible)	Day 5–7	Validated requirements, prototype feedback reports

3.3 Risk Mitigation

Risk	Mitigation Strategy
Government regulator unavailability	Use published labor laws and policies as primary source; note limitation.
Low survey response from job seekers	Offer small incentive (e.g., priority notification for new jobs).
Conflicting requirements (e.g., employer wants less data, regulator wants more)	Use MoSCoW prioritisation workshop with group vote; document conflict.

Incomplete responses from surveys	Use follow-up phone calls or SMS reminders for incomplete submissions.

4. System Architecture (MVC Compliance – Mandatory per Exam)

The system strictly follows **Laravel's MVC pattern**:

Component	Laravel Implementation	Purpose
Model	Eloquent Models (User, Job, Application)	Data logic & relationships
View	Blade templates (.blade.php)	User interface
Controller	Resource controllers (JobController, ApplicationController)	Business logic & HTTP handling
Routes	web.php (Route::resource, Route::get/post)	URL mapping

4.1 Entity-Relationship Diagram (Textual Representation) – for Migrations

The following entities and their relationships form the core data model of the Online Job Application System.

Entities and Attributes:

Entity: User (Polymorphic base)

- Attributes: id (PK), name, email, password, role (seeker/employer/admin), phone, profile_picture, is_active, email_verified_at, remember_token, created_at, updated_at

Entity: JobSeekerProfile (extends User where role = seeker)

- Attributes: id (PK), user_id (FK), date_of_birth, gender, location, education_level, years_experience, resume_path, skills (JSON/text array), notification_preferences

Entity: EmployerProfile (extends User where role = employer)

- Attributes: id (PK), user_id (FK), company_name, company_description, company_website, industry, company_logo, tax_id (optional), verified_by_admin, verification_date

Entity: Job

- Attributes: id (PK), employer_id (FK), title, description, requirements, location, salary_min, salary_max, job_type (full-time/part-time/contract/internship), deadline, status (open/closed), created_at, updated_at

Entity: Application

- Attributes: id (PK), job_id (FK), job_seeker_id (FK), cover_letter, cv_path, status (pending/reviewed/shortlisted/rejected/hired), applied_date, updated_at, employer_notes (private)

Entity: SavedJob

- Attributes: id (PK), job_seeker_id (FK), job_id (FK), saved_date

Entity: Notification

- Attributes: id (PK), user_id (FK), type (email/sms/app), subject, message, is_read, sent_at, delivery_status (pending/sent/failed)

Entity: AuditLog (for Admin & Regulator)

- Attributes: id (PK), admin_id (FK), action, target_type (User/Job/Application), target_id, old_values (JSON), new_values (JSON), ip_address, created_at

Entity: RegulatoryReport (for Government Body)

- Attributes: id (PK), report_type, generated_by (admin_id), date_range_start, date_range_end, aggregated_data (JSON), status (draft/submitted/archived)

Key Relationships:

- A User (seeker) has **one** JobSeekerProfile (1:1)
- A User (employer) has **one** EmployerProfile (1:1)
- An EmployerProfile has **many** Job postings (1:M)
- A Job has **many** Application submissions (1:M)
- A User (seeker) has **many** Application submissions (1:M)
- A User (seeker) has **many** SavedJob (1:M)
- A User (any) has **many** Notification (1:M)
- An Admin (special user role) has **many** AuditLog entries (1:M)
- An Admin generates **many** RegulatoryReport (1:M)

4.2 Class Diagram Summary (Laravel MVC Mapping)

The system is organised into the following packages:

- **Authentication & Authorization Package:** Contains User model (with role attribute), authentication controllers (Login/Register), and middleware (IsSeeker, IsEmployer, IsAdmin).
- **Job Management Package:** Contains Job model, JobController (with full CRUD methods), and Blade views for job listing, creation, editing, and deletion.
- **Application Package:** Contains Application model, ApplicationController, and views for submitting, tracking, and managing applications.
- **Profile Package:** Contains JobSeekerProfile and EmployerProfile models, profile controllers, and edit views.
- **Notification Package:** Contains Notification model, NotificationController, and service classes that interface with Third-Party Providers (Mailgun, Twilio, etc.).
- **Admin & Compliance Package:** Contains AuditLog and RegulatoryReport models, admin controllers for user management and report generation.

5. Requirements Modelling and Analysis

5.1 Use Case Diagram – Actors and Use Cases

Actors (5 total):

1. **Job Seeker** – Registers, updates profile, searches jobs, applies, saves jobs, tracks status.
2. **Employer** – Registers company, posts/edits/deletes jobs, reviews applications, updates application status.
3. **System Administrator** – Manages users, monitors system, manages third-party API keys, generates reports.
4. **Government Regulatory Body** – Receives anonymized reports, flags discriminatory jobs, views compliance data.
5. **Third-Party Service Provider** – Provides email/SMS delivery (system-to-system, no direct user interaction).

Primary Use Cases:

Use Case ID	Use Case Name	Primary Actor(s)
UC-01	Register Account	Job Seeker, Employer
UC-02	Login to System	Job Seeker, Employer, Admin
UC-03	Update Profile	Job Seeker, Employer
UC-04	Post a Job (Create)	Employer
UC-05	Edit a Job (Update)	Employer
UC-06	Delete a Job (Delete)	Employer
UC-07	View My Jobs (Read)	Employer
UC-08	Browse & Search Jobs	Job Seeker
UC-09	Apply for a Job	Job Seeker
UC-10	Track Application Status	Job Seeker
UC-11	Review Applications (per job)	Employer
UC-12	Update Application Status	Employer
UC-13	Save Job for Later	Job Seeker
UC-14	Manage Users (Suspend/Delete)	System Administrator
UC-15	View System Analytics / Audit Log	System Administrator
UC-16	Generate Regulatory Report	System Administrator (for Government Body)
UC-17	Flag Discriminatory Job Posting	Government Regulatory Body
UC-18	Send Notification (Email/SMS)	System (triggered by UC-09, UC-12) → Third-Party Provider

5.2 System Sequence Overview (End-to-End Job Application)

A complete job application flow involving all five stakeholders:

1. **Employer** registers company (UC-01). System Administrator verifies employer (UC-14).
2. **Employer** posts a job (UC-04). System stores job with status open.

3. **Job Seeker** registers, builds profile (UC-01, UC-03).
4. **Job Seeker** searches and applies for the job (UC-09). System validates CV and cover letter.
5. **System** triggers ApplicationCreated event; calls **Third-Party Service Provider** (email/SMS) ; Employer receives notification (UC-18).
6. **Employer** reviews applications (UC-11) and updates status to "Shortlisted" (UC-12).
7. **System** triggers StatusChanged event ; **Third-Party Service Provider** sends email/SMS to Job Seeker (UC-18).
8. (Compliance) **Government Regulatory Body** requests quarterly hiring report. **System Administrator** generates anonymized report (UC-16) and shares.
9. (Oversight) **Government Regulatory Body** flags a discriminatory job posting; **System Administrator** removes it (UC-17, UC-14).

6. Scenario and Use Case Development

6.1 UC-04: Post a Job (Employer)

Field	Description
Use Case ID	UC-04
Title	Post a New Job Vacancy
Primary Actor	Employer (Company/HR)
Secondary Actors	System (database), System Administrator (for auto-compliance check)
Preconditions	Employer is logged in. Employer profile is complete (company name, contact).
Postconditions	New job record created in jobs table (status = open). Job visible to job seekers. Optional compliance flag sent to admin if job contains suspicious terms.

Main Success Scenario:

1. Employer navigates to "Post a Job" from dashboard.
2. System displays a form with fields: title, description, requirements, location, salary range, job type, deadline.
3. Employer fills all required fields and submits.
4. System validates input (title required, deadline future, salary numeric positive).
5. System creates new Job record with employer_id = current user, status = open.
6. System performs automated keyword check for discriminatory language (e.g., "only male", "age < 25"). If flagged, notifies admin.
7. System redirects to "My Jobs" page with success flash message.
8. Job immediately appears in public job listings.

Alternative Flows:

- *4a. Missing required fields:* System redisplays form with error messages.
- *5a. Deadline in past:* System rejects submission, asks for future date.
- *6a. Discriminatory language suspected:* System still posts job but flags for admin review within 24 hours. Employer receives warning message.

6.2 UC-09: Apply for a Job (Job Seeker) with Third-Party Notification

Field	Description
Use Case ID	UC-09
Title	Submit Job Application (with Notification to Employer)
Primary Actor	Job Seeker
Secondary Actors	System, Third-Party Service Provider (Email/SMS)
Preconditions	Seeker is logged in. Seeker profile has at least a CV uploaded. Job is open and deadline not passed.
Postconditions	Application record created. System calls Third-Party Service Provider API. Employer receives email/SMS notification. Seeker receives confirmation.

Main Success Scenario:

1. Seeker views a job detail page.
2. Seeker clicks "Apply Now".
3. System checks for duplicate application (same seeker, same job). If none, proceeds.
4. System displays application form (cover letter textarea, CV upload pre-filled from profile).
5. Seeker writes cover letter and optionally uploads new CV.
6. Seeker submits application.
7. System validates cover letter (min 50 characters) and CV file (PDF/DOCX, max 2MB).
8. System stores CV in storage/app/public/cvs/ with unique filename.
9. System creates Application record with status = pending.
10. **System calls Third-Party Service Provider API** to send email to employer (new application notification) and SMS to employer (optional if phone provided).
11. Third-Party Provider returns success/failure; system logs delivery status in notifications table.
12. System redirects seeker to "My Applications" with success message.

Alternative Flows:

- *3a. Already applied:* System shows "You have already applied" and disables button.

- *7a. Invalid file type/size:* System rejects with specific error message.
- *10a. Third-Party API fails (timeout/error):* System queues notification for retry (up to 3 times). Logs failure for admin review.

6.3 UC-12: Update Application Status (Employer) with Notification

Field	Description
Use Case ID	UC-12
Title	Change Application Status (e.g., Shortlist, Reject, Hire)
Primary Actor	Employer
Secondary Actors	System, Third-Party Service Provider (Email/SMS to Job Seeker)
Preconditions	Employer is logged in. Application exists for a job owned by this employer.
Postconditions	applications.status updated. System calls Third-Party Provider. Job seeker receives email/SMS about status change.

Main Success Scenario:

1. Employer views applications list for a job (UC-11).
2. Next to an applicant, employer clicks dropdown and selects new status (e.g., "Shortlisted").
3. System displays confirmation dialog: "Change status to Shortlisted? This will notify the applicant."
4. Employer confirms.
5. System updates status field in applications table.
6. **System calls Third-Party Service Provider API** to send email/SMS to job seeker: "Your application for [Job Title] has been updated to Shortlisted."
7. Third-Party provider returns success; system logs notification.
8. System refreshes page with success message.

Alternative Flows:

- *3a. Employer cancels:* No change made.
- *5a. Database update fails:* System rolls back, shows error, logs to admin.
- *6a. Notification fails (seeker has invalid email/phone):* System logs failure, displays warning to employer but status change remains.

6.4 UC-14: Manage Users (System Administrator)

Field	Description
Use Case ID	UC-14
Title	Suspend or Delete User Accounts
Primary Actor	System Administrator
Secondary Actors	None (logged internally)
Preconditions	Admin is logged in with role = admin.

Postconditions	User account suspended (cannot log in) or deleted (soft or hard). Audit log entry created.

Main Success Scenario (Suspend User):

1. Admin navigates to "User Management" dashboard.
2. Admin searches for a specific user (seeker or employer) by name/email.
3. System displays user profile summary (registration date, role, activity log).
4. Admin clicks "Suspend Account".
5. System prompts: "Enter reason for suspension (e.g., fake job posts, spam applications)."
6. Admin enters reason and confirms.
7. System updates users.is_active = false.
8. System creates entry in audit_logs table with action="suspend", reason, admin_id, timestamp.
9. System shows success message: "User suspended. They will be unable to log in."

Alternative Flows (Delete User – Hard):

- *4a. Admin clicks "Delete Account" (hard delete).*
- *5a. System warns: "This action is permanent. All jobs/applications will be removed. Proceed?"*
- *6a. Admin confirms.*
- *7a. System cascades delete to related records (Eloquent delete).*
- *8a. System logs deletion in audit_logs.*

6.5 UC-16: Generate Regulatory Report (Admin → Government Body)

Field	Description
Use Case ID	UC-16
Title	Generate Anonymized Labor Market Report for Regulatory Body
Primary Actor	System Administrator (acting on behalf of Government Regulatory Body request)
Secondary Actors	Government Regulatory Body (receives report)
Preconditions	Admin is logged in. Government Body has requested data (e.g., quarterly report).
Postconditions	Anonymized, aggregated report generated (JSON/PDF). No individual user or employer identifiable. Report stored in regulatory_reports table.

Main Success Scenario:

1. Admin navigates to "Compliance Reports" section.
2. Admin selects "Generate New Report".
3. Admin sets parameters: date range (e.g., Q1 2026), report type (e.g., "Hiring by Industry", "Salary Trends", "Application Volume").
4. System queries database, aggregating data without exposing PII:

    - Total jobs posted (by industry, job type)
    - Total applications submitted
    - Average salary range by industry
    - Top 10 job titles
    - Application-to-hire conversion rate
5. System generates JSON data and optionally a PDF summary.
6. System stores report in regulatory_reports table with status = "draft".
7. Admin reviews report, clicks "Submit to Regulator".
8. System changes status to "submitted", logs timestamp and admin_id.
9. Admin downloads report and sends via secure channel (email/encrypted drive) to Government Body contact.

Alternative Flows:

- *4a. No data in date range:* System returns empty report with note: "No jobs posted in this period."
- *7a. Admin rejects draft:* Admin adds notes, regenerates.

7. Functional Requirements Specification

Functional Requirements (Organized by Stakeholder)

ID	Requirement Description	Priority	Source Stakeholder	Rationale
Job Seeker (FR-JS)				
FR-JS-01	The system shall allow job seekers to register using email/phone and password.	Must	Job Seeker	Access control
FR-JS-02	The system shall allow job seekers to create, view, and edit their profile (education, skills, CV).	Must	Job Seeker	Data management
FR-JS-03	The system shall allow job seekers to browse and search jobs by keyword, location, job type, and salary range.	Must	Job Seeker	Usability
FR-JS-04	The system shall allow job seekers to apply for a job with a cover letter and CV upload.	Must	Job Seeker	Core functionality

FR-JS-05	The system shall prevent duplicate applications (same seeker, same job).	Must	Employer	Data integrity
FR-JS-06	The system shall allow job seekers to track the status of their applications (pending/reviewed/shortlisted/rejected/hired).	Must	Job Seeker	Transparency
FR-JS-07	The system shall allow job seekers to save jobs to a "Saved Jobs" list.	Should	Job Seeker	Convenience
Employer (FR-EM)				
FR-EM-01	The system shall allow employers to register their company and create an employer profile.	Must	Employer	Onboarding
FR-EM-02	The system shall allow employers to perform full CRUD operations on job postings (Create, Read, Update, Delete).	Must	Employer	Core functionality
FR-EM-03	The system shall allow employers to view all applications per job with filtering by status.	Must	Employer	Workflow
FR-EM-04	The system shall allow employers to update application status (pending → reviewed → shortlisted → rejected → hired).	Must	Employer	Workflow
FR-EM-05	The system shall provide employers with basic analytics: views per job, applications per job.	Should	Employer	Decision support
FR-EM-06	The system shall allow employers to download applications data as CSV.	Could	Employer	Record keeping
System Administrator (FR-ADMIN)				
FR-ADMIN-01	The system shall allow administrators to view, suspend, or delete user accounts (seekers and employers).	Must	System Admin	Moderation
FR-ADMIN-02	The system shall maintain an audit log of all admin actions (suspend, delete, flag).	Must	System Admin	Accountability
FR-ADMIN-03	The system shall allow administrators to view system health metrics (uptime, error logs).	Must	System Admin	Maintenance
FR-ADMIN-04	The system shall allow administrators to manage third-party API keys (email, SMS).	Must	System Admin	Integration
Government Regulatory				

Body (FR-GOV)				
FR-GOV-01	The system shall allow administrators to generate anonymized, aggregated labor market reports for the regulatory body.	Should	Government Body	Compliance
FR-GOV-02	The system shall provide a mechanism for the regulatory body to flag discriminatory or illegal job postings (via admin mediation).	Could	Government Body	Enforcement
FR-GOV-03	The system shall retain regulatory reports for a minimum of 3 years.	Should	Government Body	Audit trail
Third-Party Service Provider (FR-TP)				
FR-TP-01	The system shall integrate with at least one third-party email service provider (e.g., Mailgun, SendGrid) via API.	Must	System + TP	Notifications
FR-TP-02	The system shall integrate with at least one third-party SMS provider (e.g., Twilio, Africa's Talking) via API.	Should	System + TP	Notifications
FR-TP-03	The system shall implement retry logic (3 attempts, exponential backoff) for failed notification deliveries.	Must	System Admin	Reliability
FR-TP-04	The system shall log all notification delivery attempts and outcomes.	Must	System Admin	Troubleshooting

8. Non-Functional Requirements Specification

ID	Category	Requirement	Measurable Target	Source Stakeholder
NFR-01	Performance	Job listing page loads within 2 seconds under normal load (100 concurrent users).	≤ 2 seconds (95th percentile)	Job Seeker, Employer
NFR-02	Performance	Application submission completes within 3 seconds (including file upload).	≤ 3 seconds	Job Seeker
NFR-03	Security	All passwords stored using bcrypt (cost factor 10).	100% of passwords hashed	System Admin

NFR-04	Security	File uploads (CVs) validated for type (PDF/DOCX) and size (max 2MB).	100% of uploads validated	System Admin, Employer
NFR-05	Security	All database queries use Eloquent ORM (prevents SQL injection).	0 raw SQL queries in production	System Admin
NFR-06	Security	API keys for third-party providers stored encrypted (Laravel encrypt() or env).	No plaintext keys in database	System Admin, Third-Party
NFR-07	Usability	User interface built with Laravel Blade templates and responsive CSS (Bootstrap/Tailwind).	Works on mobile (320px width) and desktop	Job Seeker, Employer
NFR-08	Usability	Forms provide clear, inline validation error messages.	100% of validation failures show user-friendly message	Job Seeker
NFR-09	Availability	System available during core hours (8am – 8pm EAT) with 99% uptime.	≤ 7.2 hours downtime/month	Employer, Admin
NFR-10	Reliability	Email notifications delivered within 2 minutes of trigger.	≥ 95% delivered within 2 minutes	Employer, Job Seeker
NFR-11	Compliance	System anonymizes all reports for government body (no PII: name, email, phone, NIN).	0 PII fields in regulatory reports	Government Body
NFR-12	Maintainability	Code follows Laravel MVC pattern (Models, Views, Controllers separate).	Code review checklist passed	System Admin

9. Technical Specifications (Exam Alignment)

Component	Technology
Backend Framework	Laravel 10 or 11
PHP Version	8.1+
Database	MySQL (Migrations & Seeders)
Frontend	Blade templates + Bootstrap 5 / Tailwind CSS
Authentication	Laravel Breeze / Jetstream
File Storage	Laravel Filesystem (public disk for CVs)
Routing	web.php with named routes
Email Service	Mailgun / SendGrid API
SMS Service	Twilio / Africa's Talking API

10. Requirements Engineering Roadmap Proposal

This section presents a **phased workflow** describing the realistic steps in requirements engineering that guide the development of the Online Job Application System, aligned with the five stakeholders.

Phase 1: Inception & Stakeholder Engagement (Day 1)

Objective: Establish project scope, identify the five stakeholders, and build engagement structures (even if simulated for the exam).

Activities:

- Conduct stakeholder identification using power-interest analysis (see Section 1.3).
- Hold a project kickoff meeting with **simulated representatives** for all five stakeholders.
- Define the project vision and scope boundaries (no payment, no AI matching, but do include third-party notification).
- Establish communication channels (email, WhatsApp group) for team members.

Deliverables: Stakeholder register (5 entries), project charter, team communication plan.

Phase 2: Requirements Elicitation (Day 1–3)

Objective: Systematically gather requirements from all five stakeholder groups using diverse techniques.

Activities:

- Conduct **structured interviews** (simulated or real) with an HR manager (Employer), a System Admin professional, and a Labor Officer (Government Body).
- Organise a **focus group workshop** with 3–5 job seekers (fellow students).
- Distribute **surveys** to job seekers (target 20+ responses via Google Forms).
- Perform **document analysis** of Uganda Employment Act (2006), Data Protection Act (2019), and third-party API docs (Twilio, Mailgun).
- Conduct **observation** of an existing job portal (BrighterMonday, Fuzu) for workflow understanding.

Deliverables: Raw requirements inventory, interview transcripts (summarized), survey analysis report, regulatory requirements matrix.

Phase 3: Requirements Analysis & Modelling (Day 3–4)

Objective: Organise, model, and prioritise the gathered requirements.

Activities:

- Classify requirements into functional (by stakeholder) and non-functional (see Section 5).
- Develop structural models (ER diagram, class diagram summary) – see Section 3.1.
- Create behavioural models (use case diagram, sequence overview) – see Section 3.2.
- Apply **MoSCoW prioritisation** with group input (Must/Should/Could/Won't).

- Identify and resolve conflicting requirements (e.g., Employer wants minimal data, Government wants more reporting). Use negotiation workshop.

Deliverables: Requirements models (ERD, use case diagram), prioritised requirements list (Section 5), conflict resolution log.

Phase 4: Requirements Specification (Day 4–5)

Objective: Produce a formal, traceable Software Requirements Specification (SRS) document (this document).

Activities:

- Write the SRS following IEEE 830 / ISO 29148 standards as adapted here.
- Assign unique identifiers to each requirement (FR-JS-xx, FR-EM-xx, FR-ADMIN-xx, FR-GOV-xx, FR-TP-xx, NFR-xx).
- Create a **Requirements Traceability Matrix** (RTM) linking each requirement to a stakeholder and to a use case.
- Develop detailed use case narratives (Section 4) with pre/post-conditions and exception flows.

Deliverables: Software Requirements Specification (this complete document), Requirements Traceability Matrix (Appendix A).

Phase 5: Requirements Validation (Day 5–7)

Objective: Ensure the specified requirements are correct, complete, consistent, and feasible within exam constraints.

Activities:

- Conduct a formal **requirements review** (walkthrough) with all group members playing stakeholder roles.
- Build **low-fidelity Blade prototypes** for key workflows (job post, apply, review applications, admin suspend user).
- Perform **requirements testing**: verify that each requirement is testable and has defined acceptance criteria.
- Use a checklist to check for completeness, ambiguity, and consistency against the exam PDF.
- **Validate third-party integration assumptions** (e.g., using dummy API keys for Mailgun/Twilio in development).

Deliverables: Validation report, updated SRS, prototype feedback summary, third-party integration test log.

Phase 6: Requirements Management (Ongoing)

Objective: Establish processes for managing changes to requirements throughout the system's lifecycle (up to presentation day).

Activities:

- Implement a **Change Control Board (CCB)** : group lead + one backend + one frontend member.
- Define a formal change request process: submission → impact analysis → review → vote → decision → implementation.
- Maintain version control of all requirements documents using Git (GitHub).
- Conduct **weekly requirements audits** (every Monday) to ensure alignment with evolving implementation and the exam deadline.

Deliverables: Change management process document, version-controlled SRS (v1.0, v1.1, etc.), audit checklists.

11. Presentation Readiness (Per Exam Requirement)

During the live demo (21–22 May 2026), the group will demonstrate:

1. **The problem being solved** – Centralized, digital job application system
2. **Key features** – Full CRUD, authentication, role-based dashboards, third-party notifications
3. **How Laravel implements MVC** – Show routes, controller, model, blade file
4. **Live demo** – Employer posts a job → Seeker applies → Employer views application → Third-party notification sent
5. **Compliance demonstration** – Admin generates anonymized report (simulated for Government Body)

All group members will participate.

12. Conclusion

This Requirements Engineering Document provides a complete, traceable, and stakeholder-driven specification for the **Online Job Application System** as required for the BSE2207 Practical Final Exam. It strictly adheres to the five-stakeholder scope (Job Seeker, Employer, System Administrator, Government Regulatory Body, Third-Party Service Provider) and excludes any reference to a "Platform Owner" evaluator role.

The document follows the structure and depth of professional RE deliverables, including stakeholder analysis, elicitation planning, structural/behavioural modelling, detailed use cases, functional/non-functional requirements tables, and a phased RE roadmap. It is now ready for use as the baseline for Laravel implementation and for the final presentation on 21–22 May 2026.

13 Appendices

Appendix A: Requirements Traceability Matrix (Sample)

Requirement ID	Source Stakeholder	Related Use Case(s)	Test Case Status
FR-JS-01	Job Seeker	UC-01	Planned
FR-EM-02	Employer	UC-04, UC-05, UC-06, UC-07	Planned
FR-ADMIN-01	System Administrator	UC-14	Planned
FR-GOV-01	Government Regulatory Body	UC-16	Planned
FR-TP-01	Third-Party Provider	UC-09, UC-12, UC-18	Planned
NFR-01	Job Seeker, Employer	UC-08	Performance test pending

Appendix B: Glossary

Term	Definition
CRUD	Create, Read, Update, Delete
MVC	Model-View-Controller
NIN	National Identification Number (not used in this system)
PII	Personally Identifiable Information
SRS	Software Requirements Specification
MoSCoW	Must, Should, Could, Won't prioritisation

Appendix C: Team Roles & Responsibilities

Member	**Role**	**Responsibility**
Kayeera Nathan	Lead Developer	Stakeholder analysis, Requirements gathering, SRS documentation, Roadmap planning, Requirements traceability, MVC system design, Module integration, System validation, hosting and Git-hub repo management
Namuganza Isabella	Frontend Developer	Blade templates, UI consistency
Nyonyozi Maria Lisa	Database Designer	Migrations, Eloquent relationships, Seeders
Ssali Peter Joshua	Backend Developer	Controllers, CRUD logic, Routing
Emuduko Benard	Security & Auth	Authentication, Validation, CSRF, Third-party APIs
Ampumuza Aijuka	QA & Presenter	Testing, Bug fixing, Live demo coordination

Prepared by Group D (Evening)
Date: May 2026